

CSA5801 - DEVSECOPS Engineering and Application Security

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VERY SIMPLE R PROGRAMS

Rule:

Static / workflow / design / modelling -> DiagrammeR

Dynamic / simulation / analysis / results -> dygraphs

#

=====

Install once:

install.packages("DiagrammeR")

install.packages("dygraphs")

install.packages("xts")

library(DiagrammeR)

library(dygraphs) library(xts)

Installation and Configuration of Git, GitHub, RStudio and

```
# -----
# EXPERIMENT 1
# Required DevSecOps Tools
# Type: Static -> Diagrammer
# -----

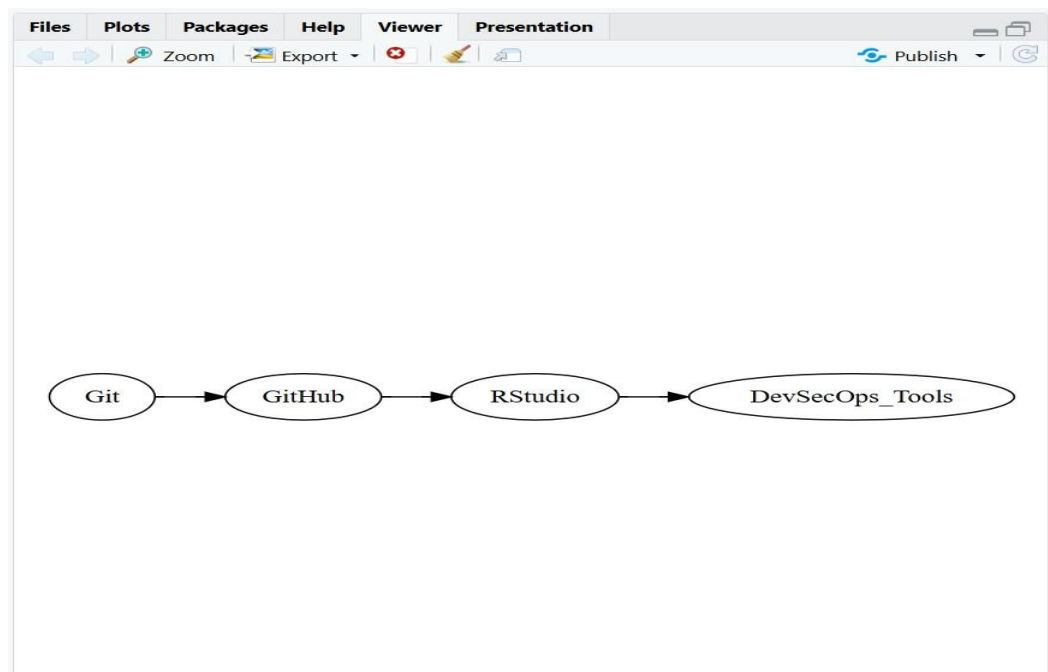
grViz(" digraph G { rankdir=LR

    Git -> GitHub -> RStudio -> DevSecOps_Tools

}

")
```

Output:



```
# Visualization of the DevSecOps Lifecycle using Workflow Modelling
# Type: Static -> Diagrammer
# -----

grViz(" digraph G { rankdir=LR

    Plan -> Code -> Build -> Test -> Deploy -> Monitor -> Plan

}

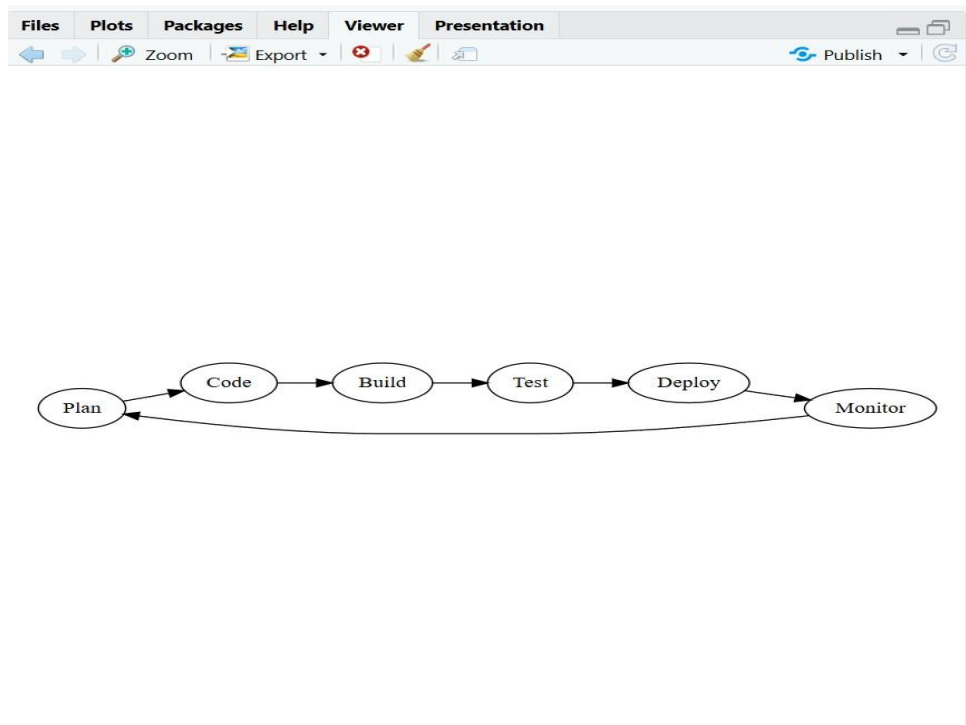
")
```

```
# -----
```

```
# EXPERIMENT 2
```

```
")
```

Output:



```
# Comparative Analysis of DevOps and DevSecOps using Synthetic Data
```

```
# Type: Dynamic -> dygraph
```

```
# ----- d <-
```

```
1:10 x <- xts(cbind(
```

```
  DevOps = c(60,62,65,63,68,70,72,71,75,76),
```

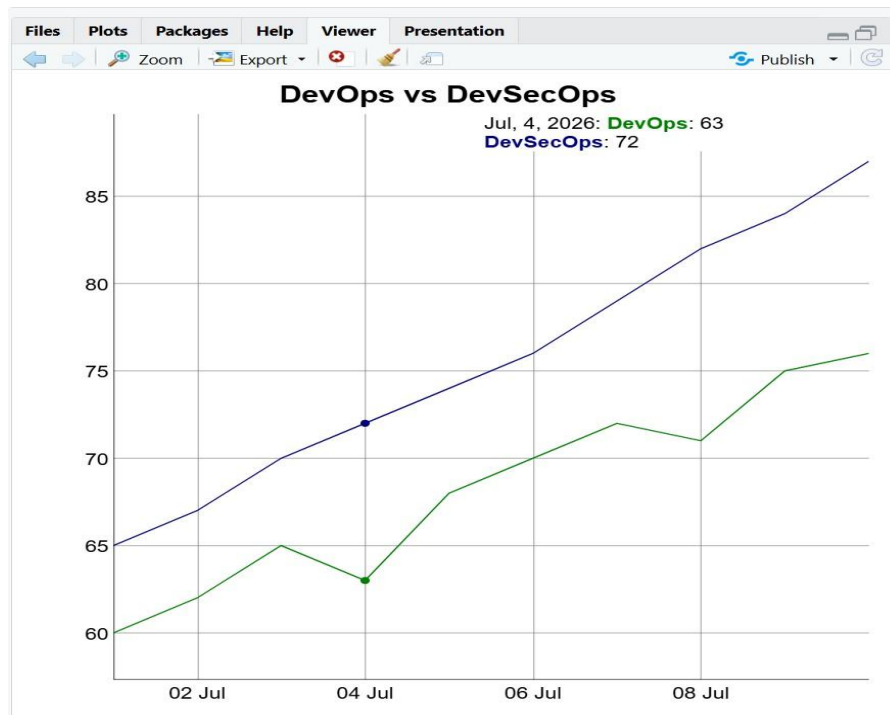
```
  DevSecOps = c(65,67,70,72,74,76,79,82,84,87) ),
```

```
order.by=as.Date("2026-07-01")+d-1) dygraph(x,
```

```
main="DevOps vs DevSecOps")
```

Output:

EXPERIMENT 3



Security Risk Assessment and Risk Matrix Simulation

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

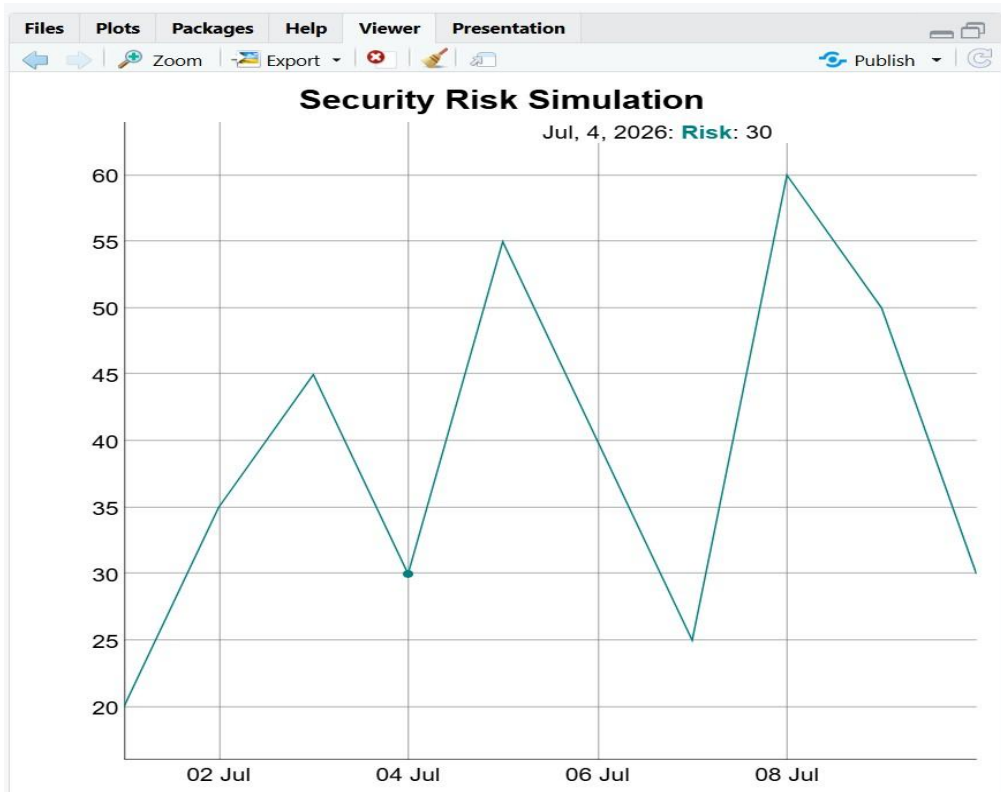
Risk = c(20,35,45,30,55,40,25,60,50,30)),

order.by=as.Date("2026-07-01")+0:9)

dygraph(x, main="Security Risk Simulation")

Output:

EXPERIMENT 4



```
# -----
```

```
# EXPERIMENT 5
```

```
# Threat Propagation Simulation and Attack Surface Analysis
```

```
# Type: Dynamic -> dygraph
```

```
# ----- x <-
```

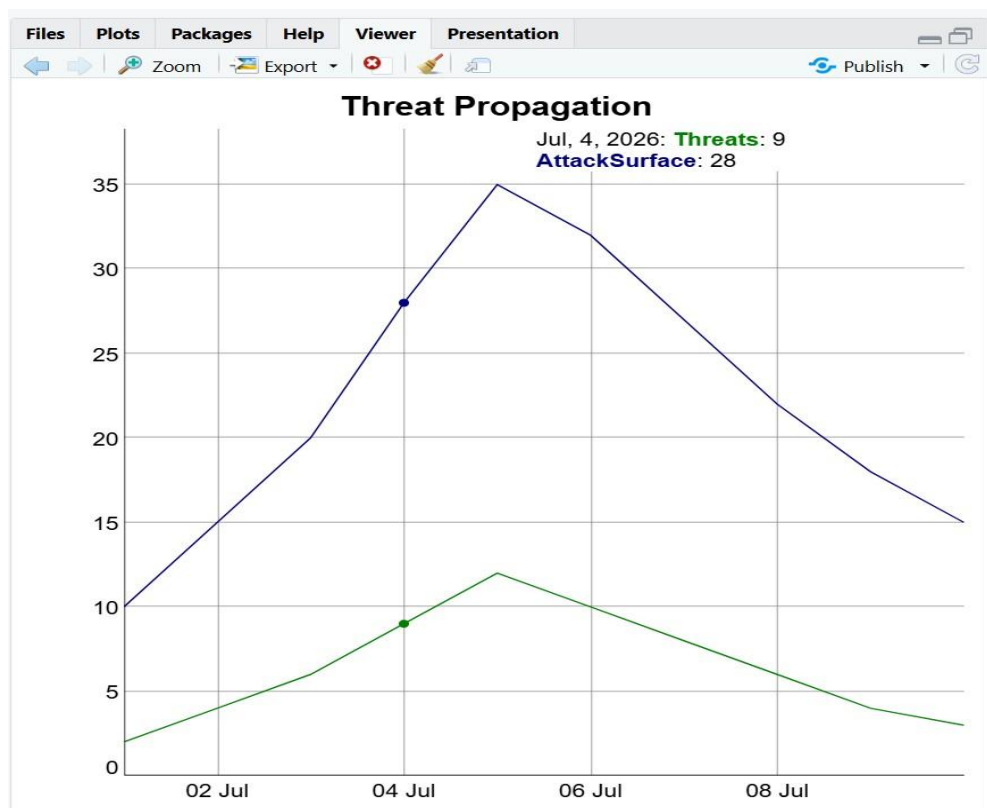
```
xts(cbind(
```

```
  Threats = c(2,4,6,9,12,10,8,6,4,3),
```

```
  AttackSurface = c(10,15,20,28,35,32,27,22,18,15) ),
```

```
order.by=as.Date("2026-07-01")+0:9) dygraph(x,
```

```
main="Threat Propagation") Output:
```



```
# Design and Simulation of Shift-Left Security Workflow
```

```
# Type: Static -> DiagrammeR
```

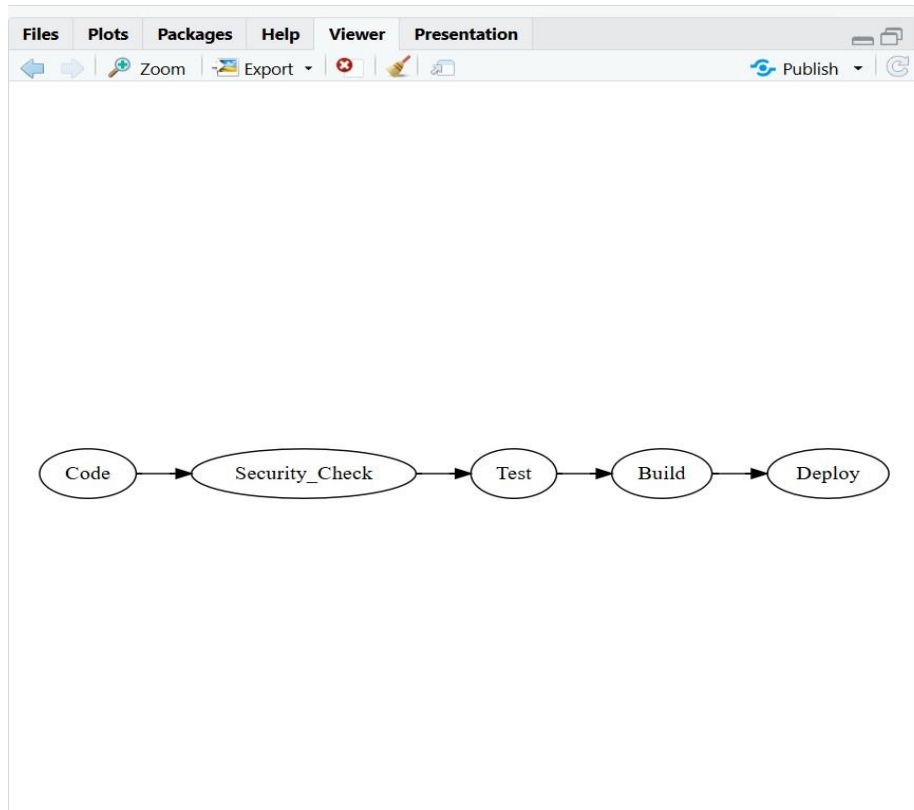
```
# -----
```

```
grViz(" digraph G { rankdir=LR
```

```
  Code -> Security_Check -> Test -> Build -> Deploy
```

```
# -----
# EXPERIMENT 6
}
")
```

Output:



```
# Static Application Security Testing (SAST) and Security Verification
```

```
# Type: Dynamic -> dygraph
```

```
# ----- x <-
```

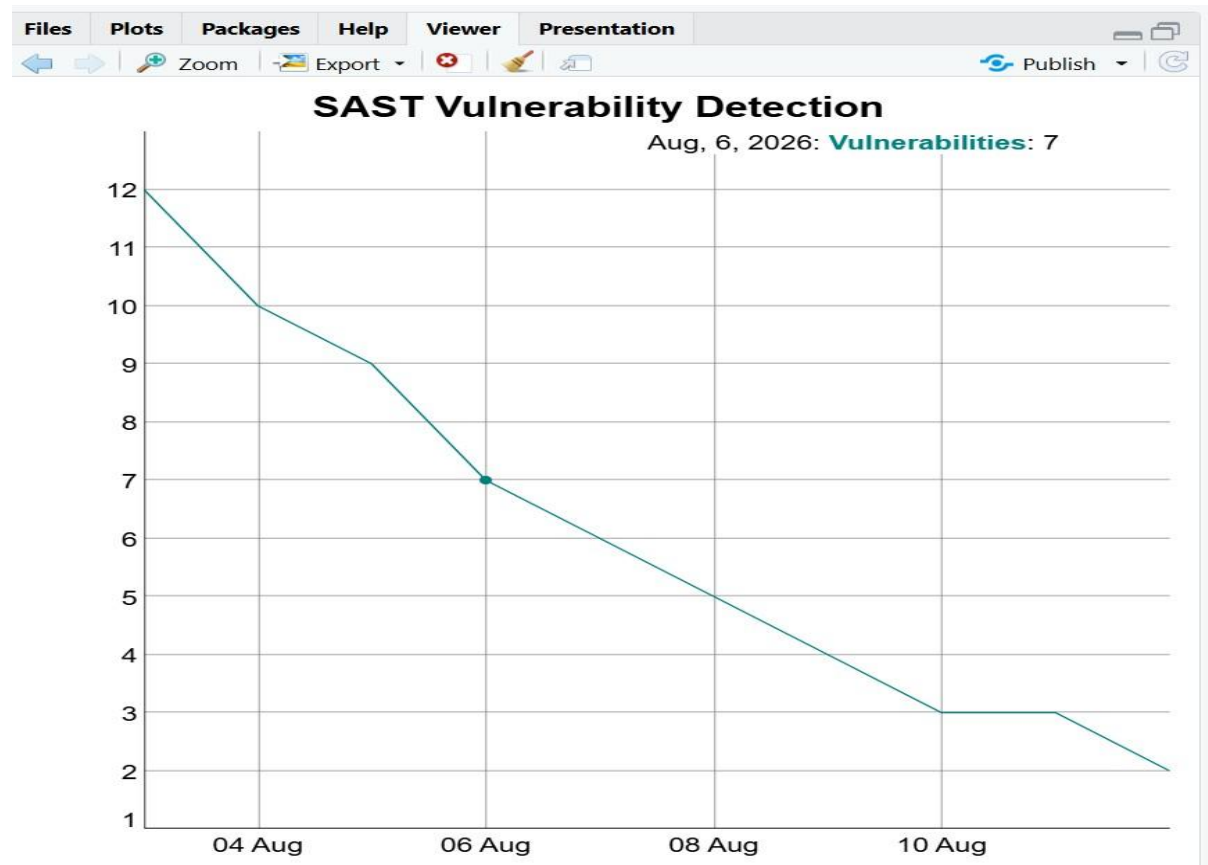
```
xts(cbind(
```

```
  Vulnerabilities = c(12,10,9,7,6,5,4,3,3,2) ),
```

```
order.by=as.Date("2026-08-03")+0:9) dygraph(x,
```

```
main="SAST Vulnerability Detection") Output:
```

EXPERIMENT 7



Dynamic Application Security Testing (DAST)

Type: Dynamic -> dygraph

----- x <-

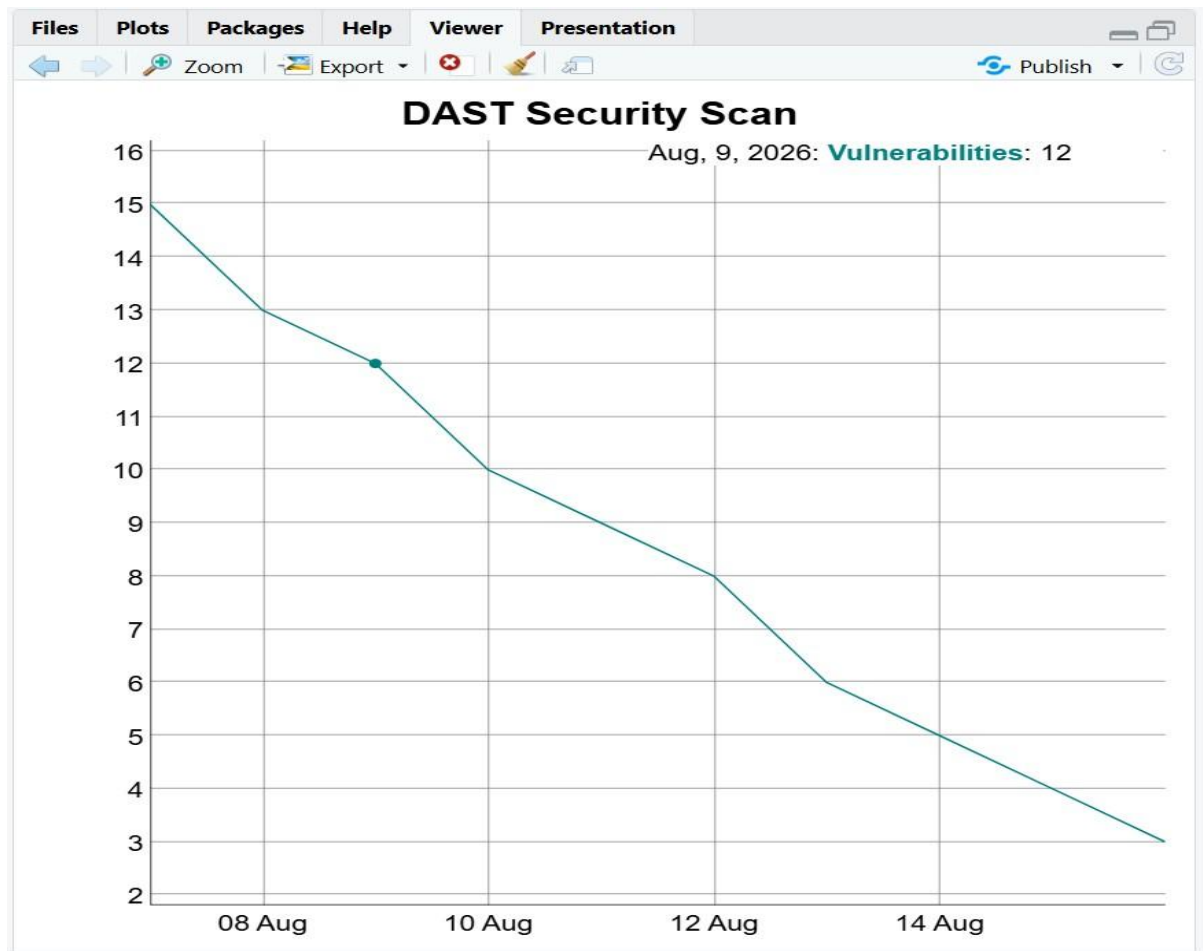
xts(cbind(

Vulnerabilities = c(15,13,12,10,9,8,6,5,4,3)),

order.by=as.Date("2026-08-07")+0:9)

dygraph(x, main="DAST Security Scan") Output:

EXPERIMENT 8



Software Composition Analysis (SCA)

Type: Dynamic -> dygraph

----- x <-

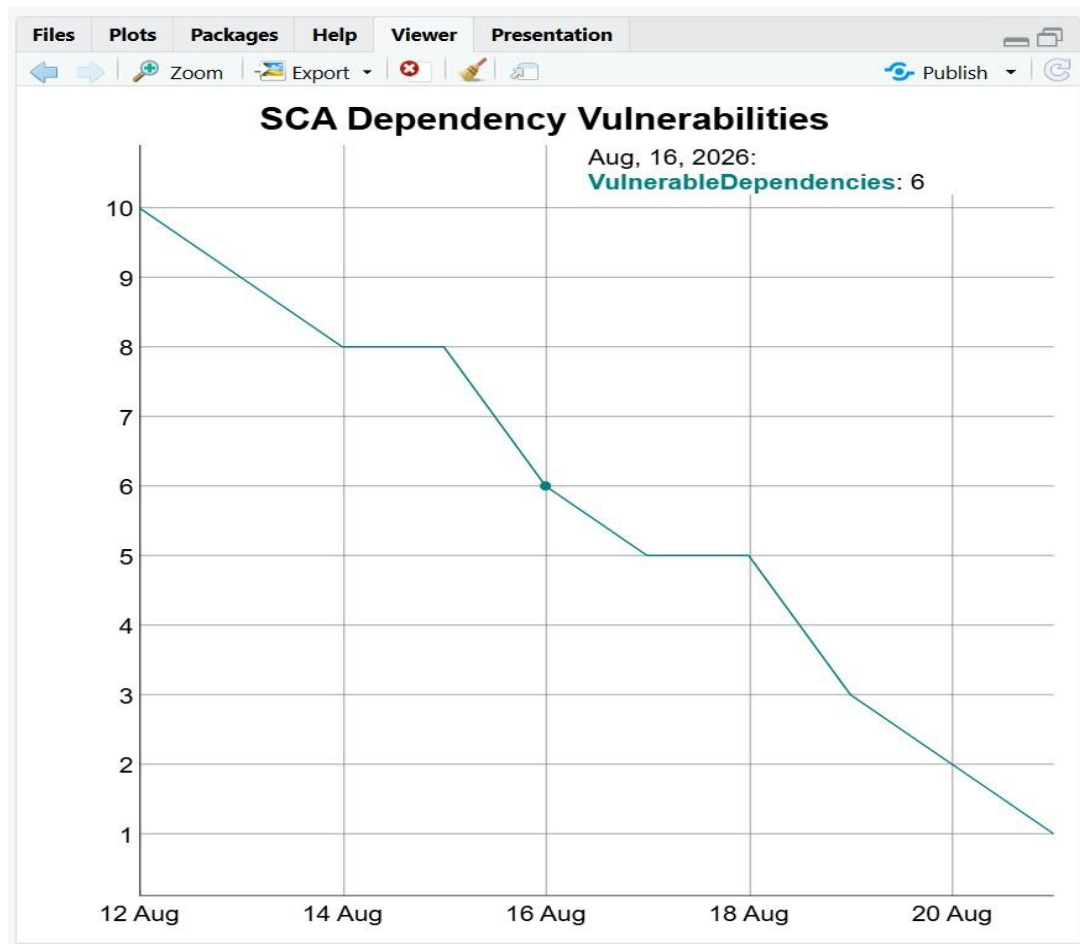
xts(cbind(

VulnerableDependencies = c(10,9,8,8,6,5,5,3,2,1)

), order.by=as.Date("2026-08-12")+0:9) dygraph(x,

main="SCA Dependency Vulnerabilities") Output:

EXPERIMENT 9



Vulnerability Trend Analysis and Security Reporting

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

 Critical = c(8,7,6,5,4,3,3,2,1,1),

 High = c(12,11,10,9,8,7,6,5,4,3)),

order.by=as.Date("2026-08-17")+0:9)

dygraph(x, main="Vulnerability Trends")

Output:

EXPERIMENT 10



Secure Infrastructure Architecture Design using IaC Concepts

Type: Static -> Diagrammer

```
grViz(" digraph G { rankdir=LR
```

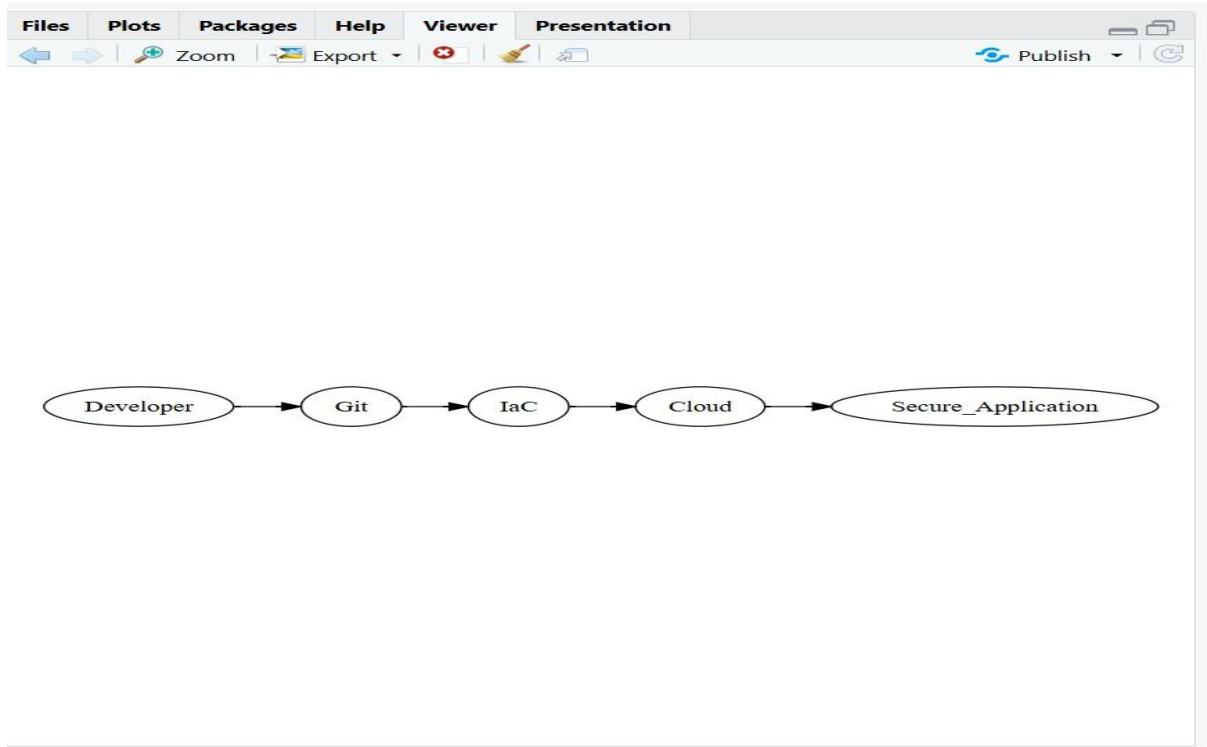
```
  Developer -> Git -> IaC -> Cloud -> Secure_Application
```

```
}
```

```
")
```

Output:

EXPERIMENT 11



Identity and Access Management (IAM) Relationship Modelling

Type: Static -> DiagrammerR

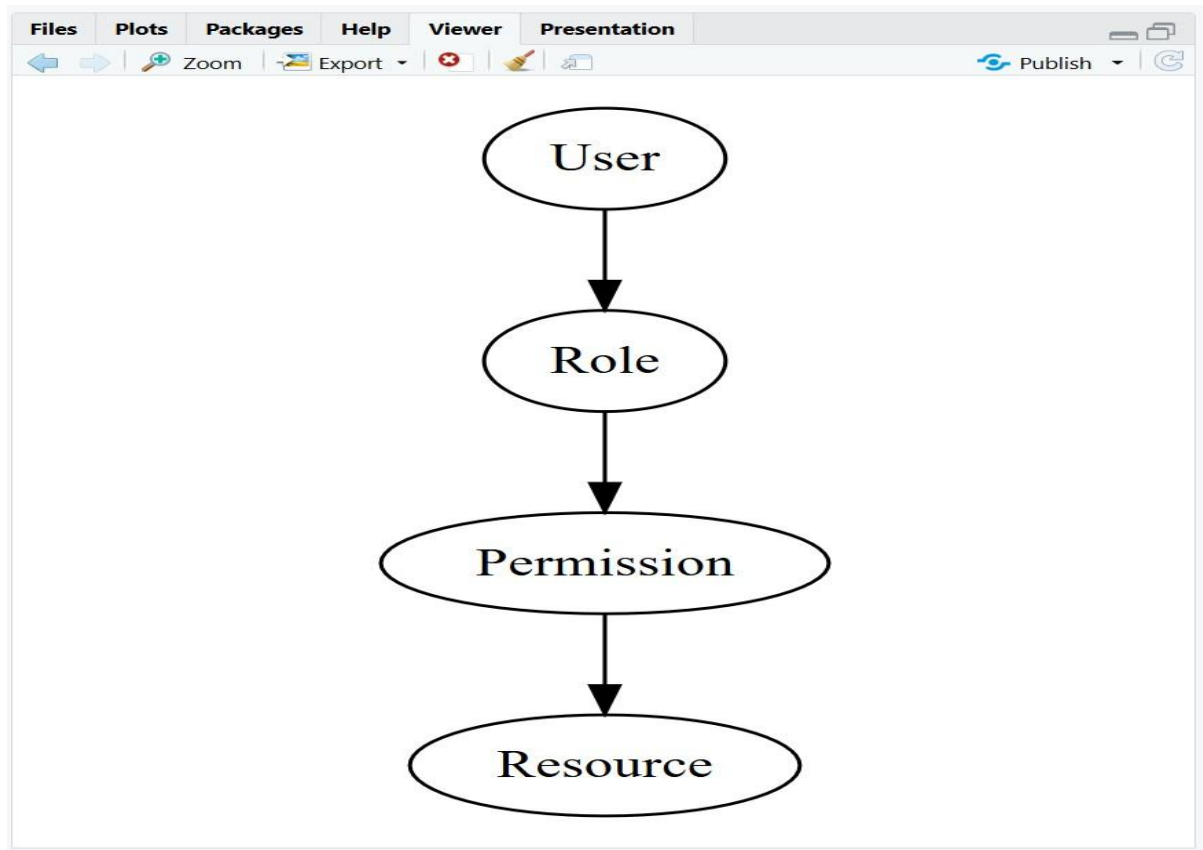
```
grViz(" digraph G {
```

```
  User -> Role -> Permission -> Resource
```

```
}
```

```
")
```

EXPERIMENT 12



GitOps Workflow Design and Secure Deployment Simulation

Type: Static -> DiagrammerR

```
grViz(" digraph G { rankdir=LR
```

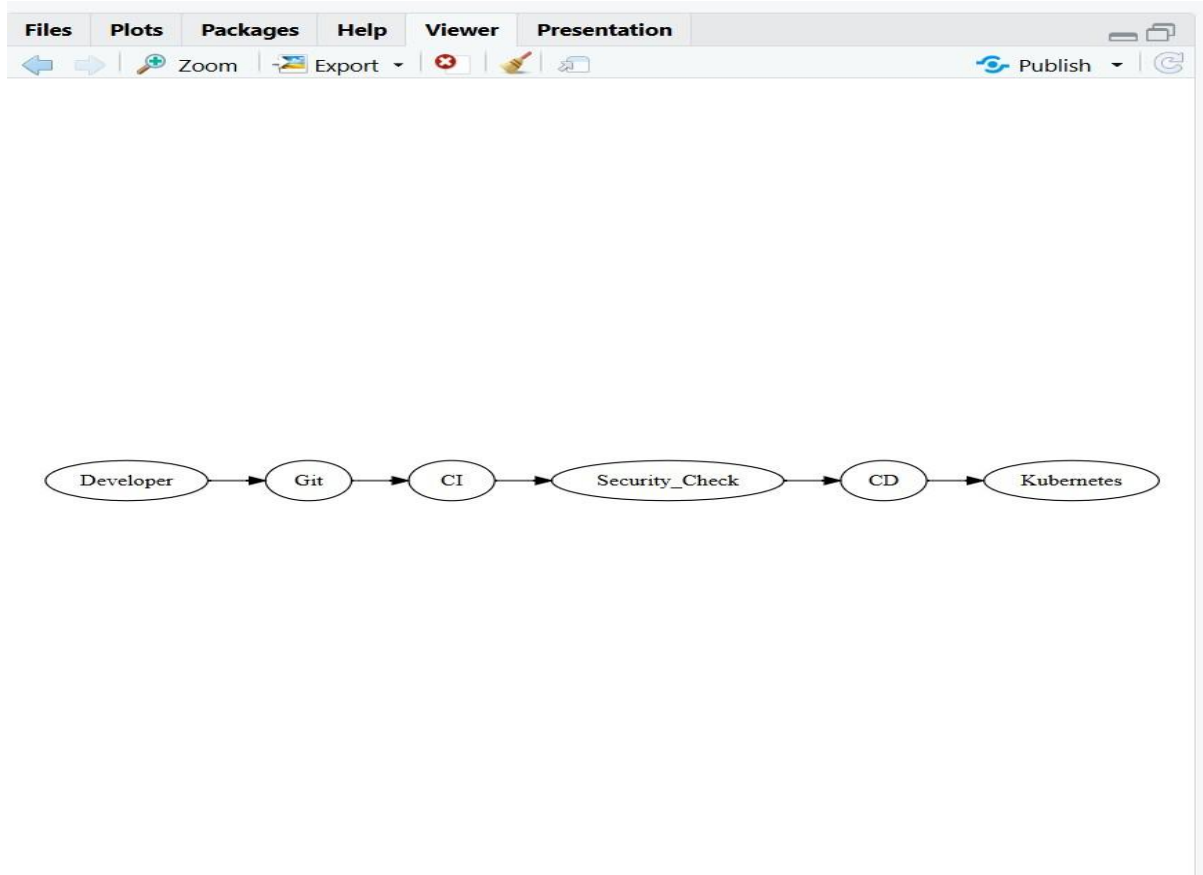
```
Developer -> Git -> CI -> Security_Check -> CD -> Kubernetes
```

```
}
```

```
")
```

Output:

EXPERIMENT 13



Container Security and Vulnerability Assessment

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

Vulnerabilities = c(14,13,11,10,8,7,6,4,3,2)),

order.by=as.Date("2026-09-04")+0:9)

dygraph(x, main="Container Vulnerability Assessment") Output:

EXPERIMENT 14



Security Log Generation, Incident Timeline Analysis and

Anomaly Detection

Type: Dynamic -> dygraph

----- x <-

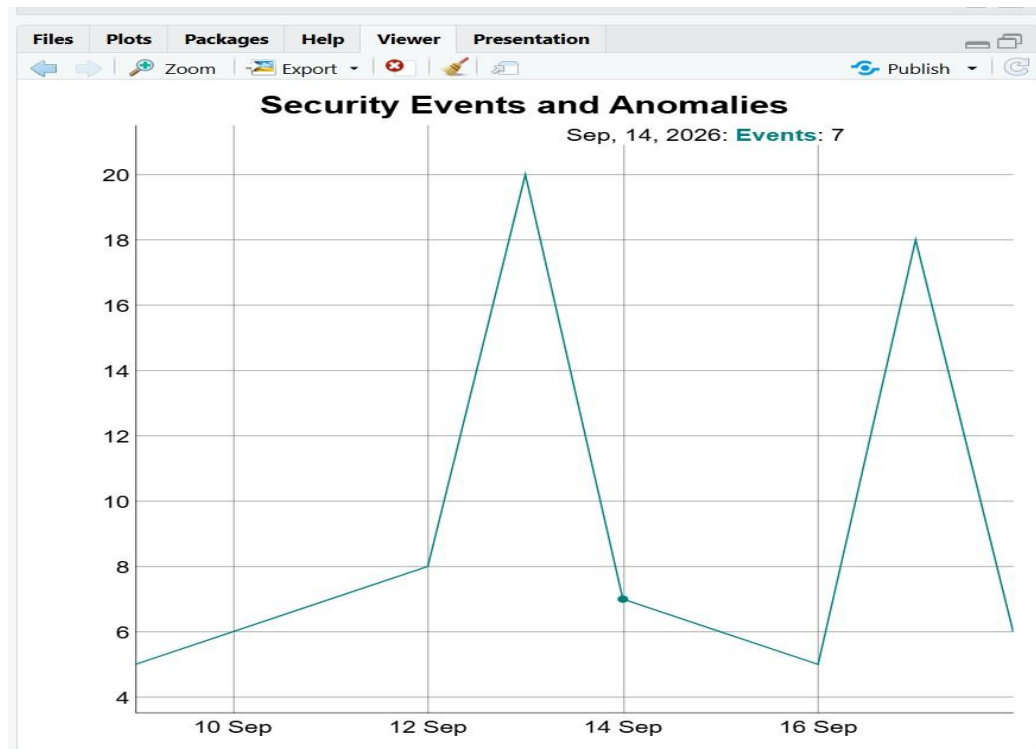
xts(cbind(

Events = c(5,6,7,8,20,7,6,5,18,6)),

order.by=as.Date("2026-09-09")+0:9) dygraph(x,

main="Security Events and Anomalies") Output:

EXPERIMENT 15



Development of a DevSecOps Security Monitoring Dashboard

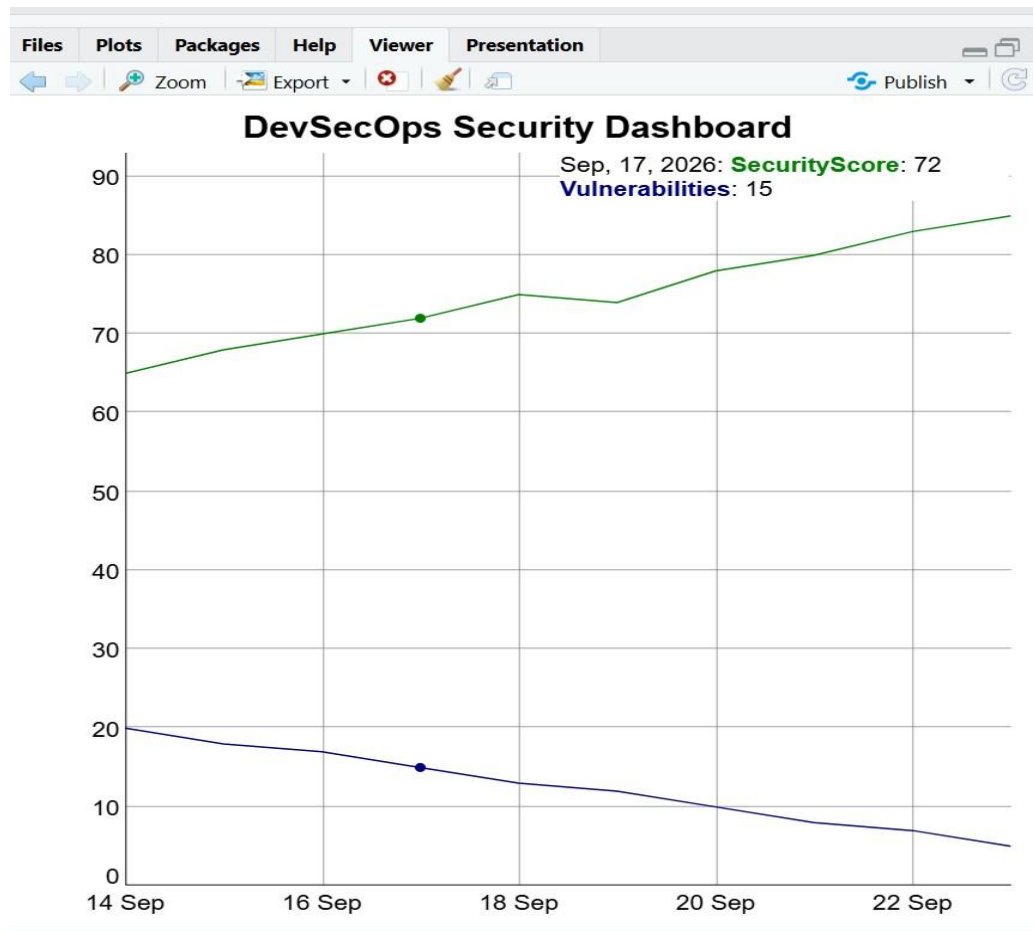
Type: Dynamic -> dygraph

----- x <-

```
xts(cbind(  
  SecurityScore = c(65,68,70,72,75,74,78,80,83,85),  
  Vulnerabilities = c(20,18,17,15,13,12,10,8,7,5) ),  
order.by=as.Date("2026-09-14")+0:9) dygraph(x,  
main="DevSecOps Security Dashboard")
```

 Output:

EXPERIMENT 16



Secure Coding Practices and Vulnerability Detection Simulation

Type: Dynamic -> dygraph

----- x <-

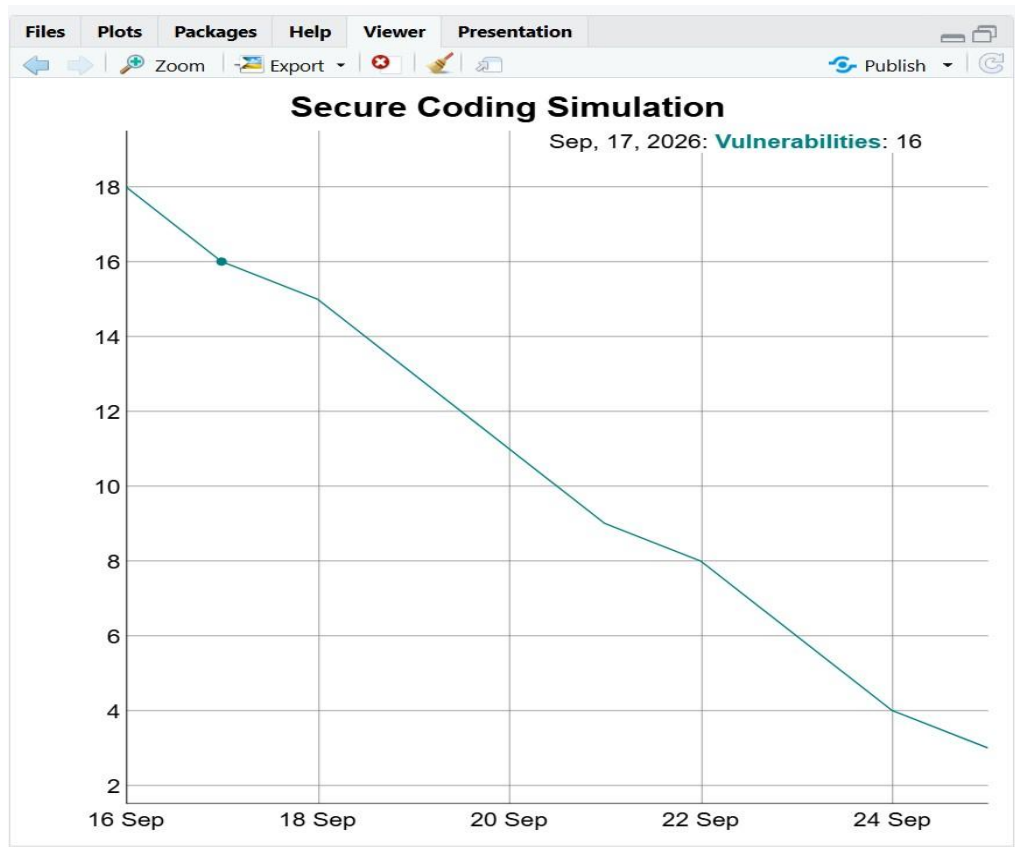
xts(cbind(

Vulnerabilities = c(18,16,15,13,11,9,8,6,4,3)),

order.by=as.Date("2026-09-16")+0:9) dygraph(x,

main="Secure Coding Simulation") Output:

EXPERIMENT 17



OWASP Top 10 Vulnerability Identification and Risk Classification

Type: Static -> DiagrammerR

```
grViz(" digraph G { rankdir=LR
```

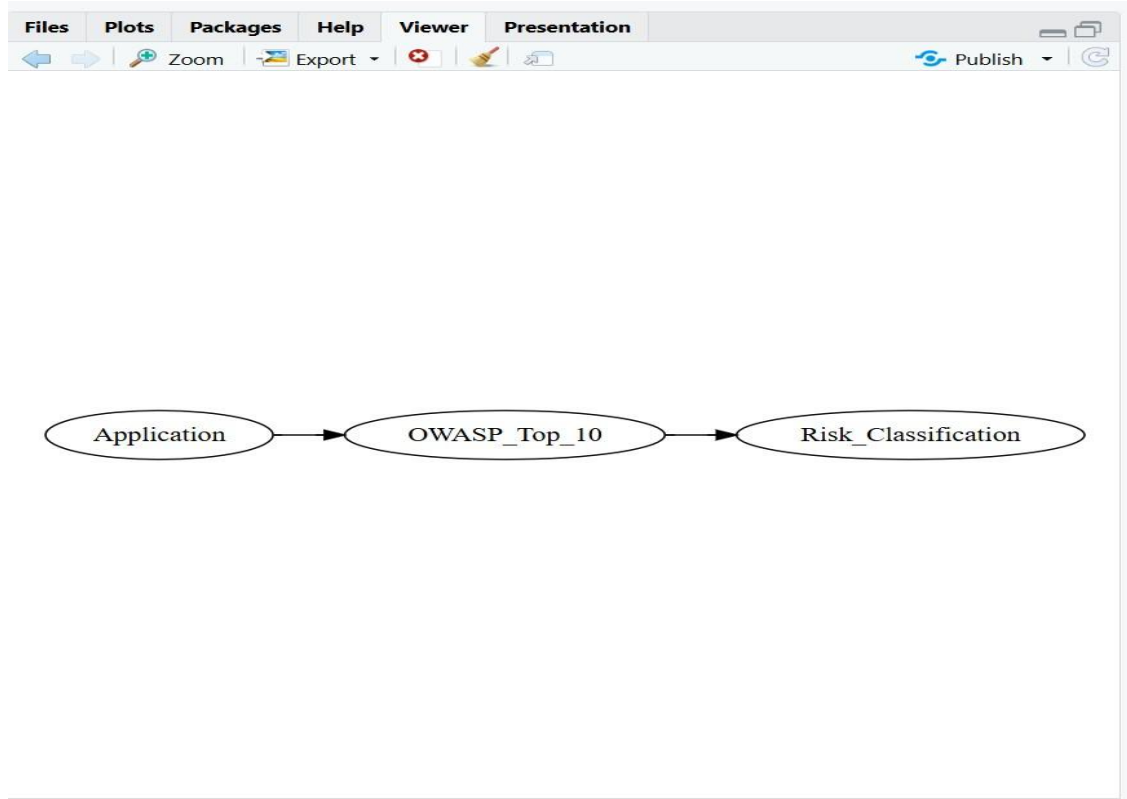
```
Application -> OWASP_Top_10 -> Risk_Classification
```

```
}
```

```
")
```

Output:

EXPERIMENT 18



Security Test Case Design and Automated Verification Simulation

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

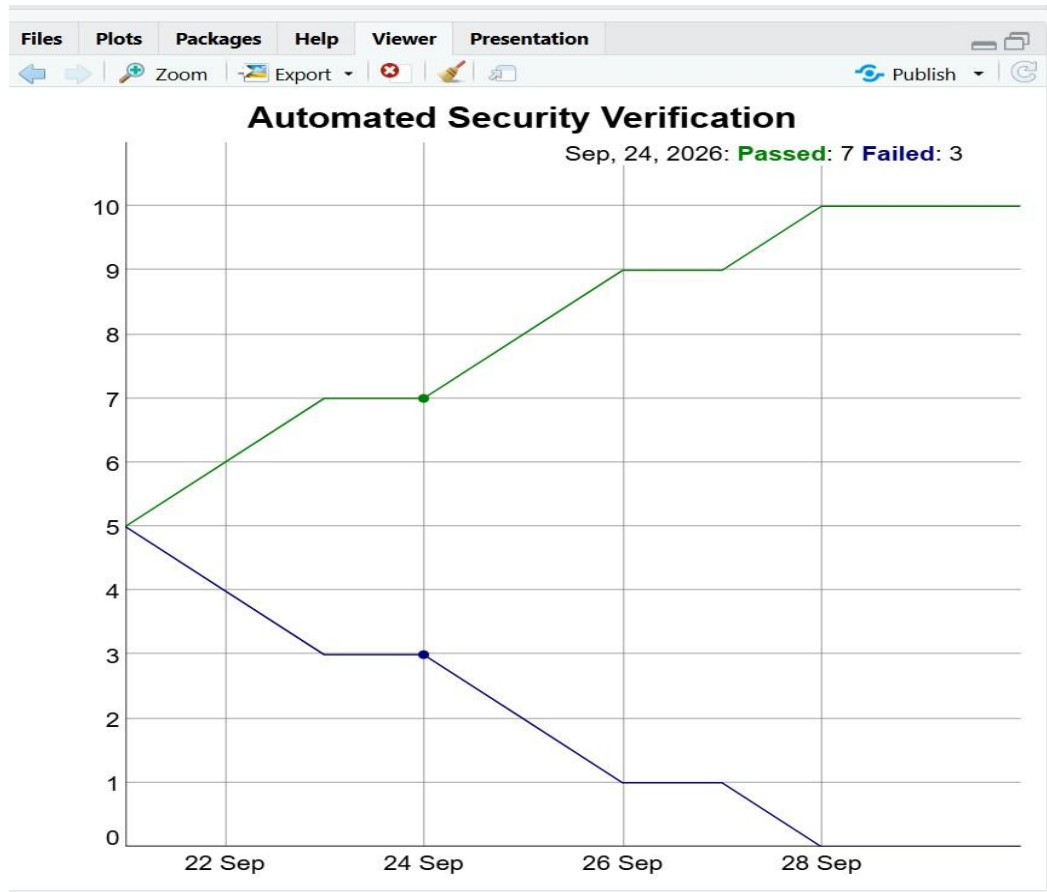
Passed = c(5,6,7,7,8,9,9,10,10,10),

Failed = c(5,4,3,3,2,1,1,0,0,0)),

order.by=as.Date("2026-09-21")+0:9) dygraph(x,

main="Automated Security Verification") Output:

EXPERIMENT 19



CI/CD Pipeline Security Verification and Quality Gate Simulation

Type: Dynamic -> dygraph

----- x <-

```
xts(cbind(
```

```
  Passed = c(3,4,5,6,7,7,8,9,9,10),
```

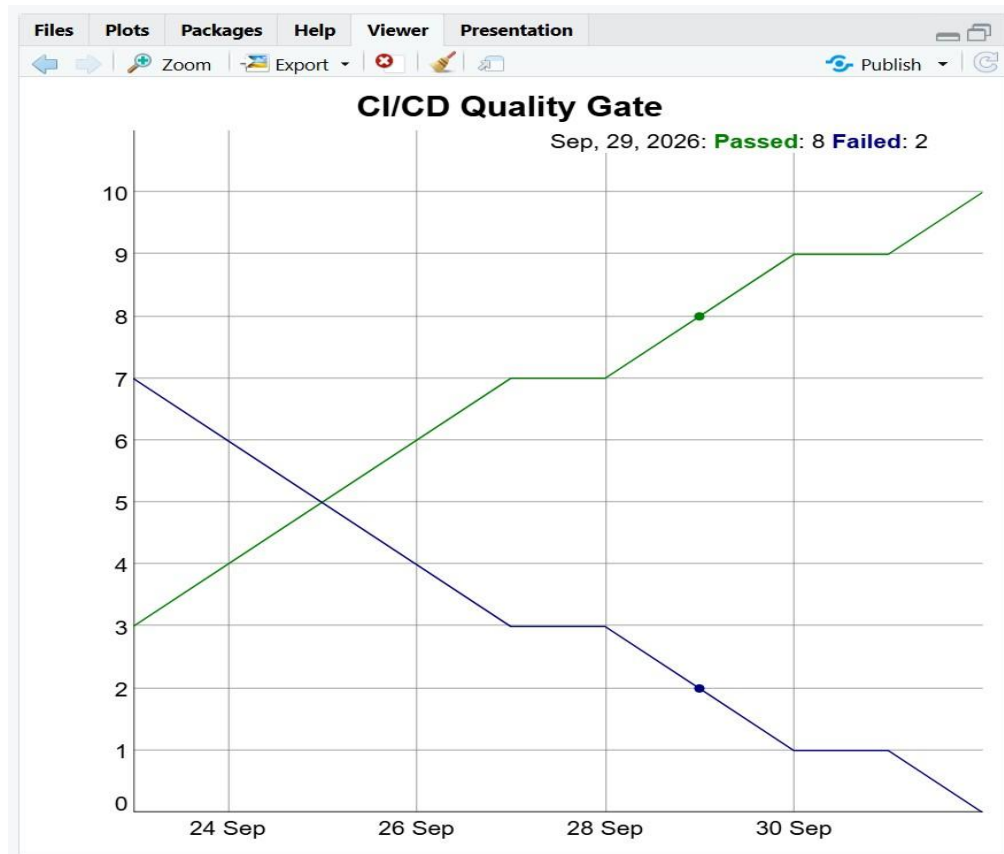
```
  Failed = c(7,6,5,4,3,3,2,1,1,0) ),
```

```
order.by=as.Date("2026-09-23")+0:9)
```

```
dygraph(x, main="CI/CD Quality Gate")
```

Output:

EXPERIMENT 20



Security Vulnerability Prioritization using CVSS-Based Scoring

Type: Dynamic -> dygraph

----- x <-

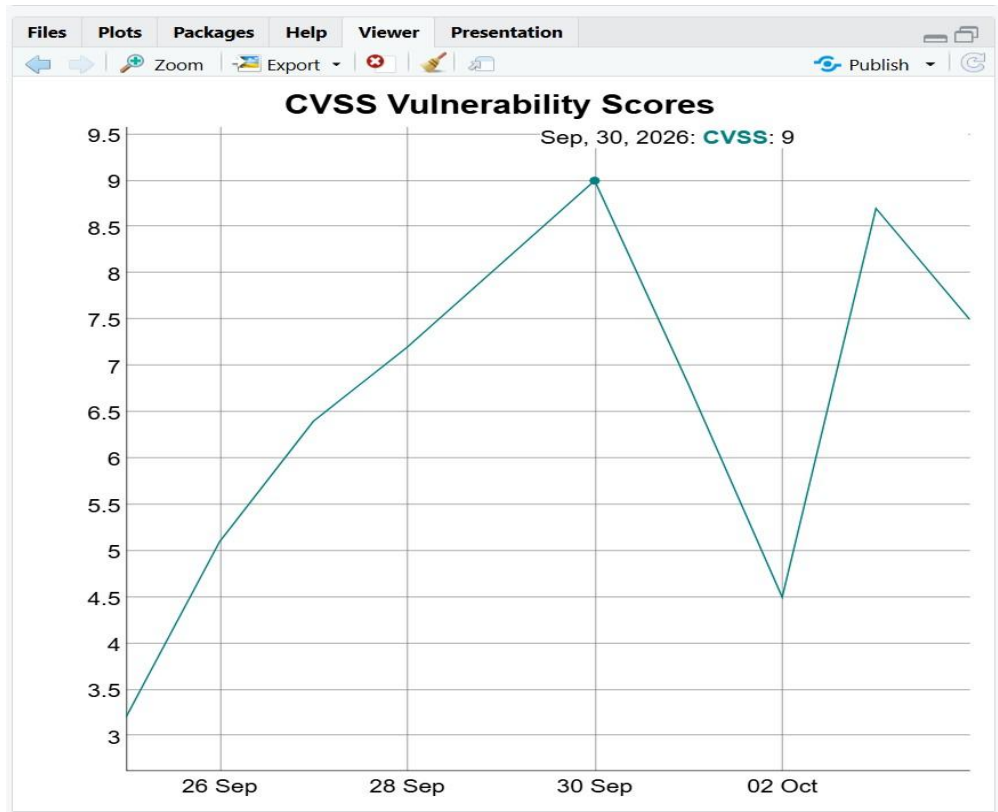
xts(cbind(

CVSS = c(3.2,5.1,6.4,7.2,8.1,9.0,6.8,4.5,8.7,7.5)),

order.by=as.Date("2026-09-25")+0:9) dygraph(x,

main="CVSS Vulnerability Scores") Output:

EXPERIMENT 21



Secrets Detection and Secure Secrets Management Simulation

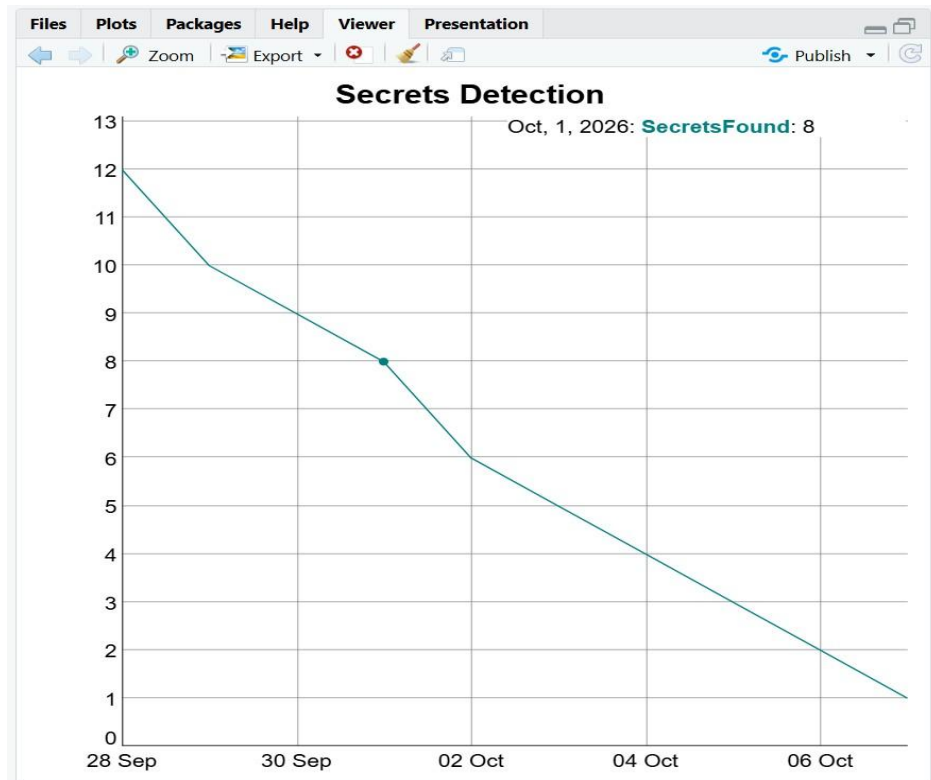
Type: Dynamic -> dygraph

----- x <-

```
xts(cbind(
  SecretsFound = c(12,10,9,8,6,5,4,3,2,1) ),
order.by=as.Date("2026-09-28")+0:9)
dygraph(x, main="Secrets Detection")
```

Output:

EXPERIMENT 22



API Security Testing and Vulnerability Analysis

Type: Dynamic -> dygraph

----- x <-

```
xts(cbind(
```

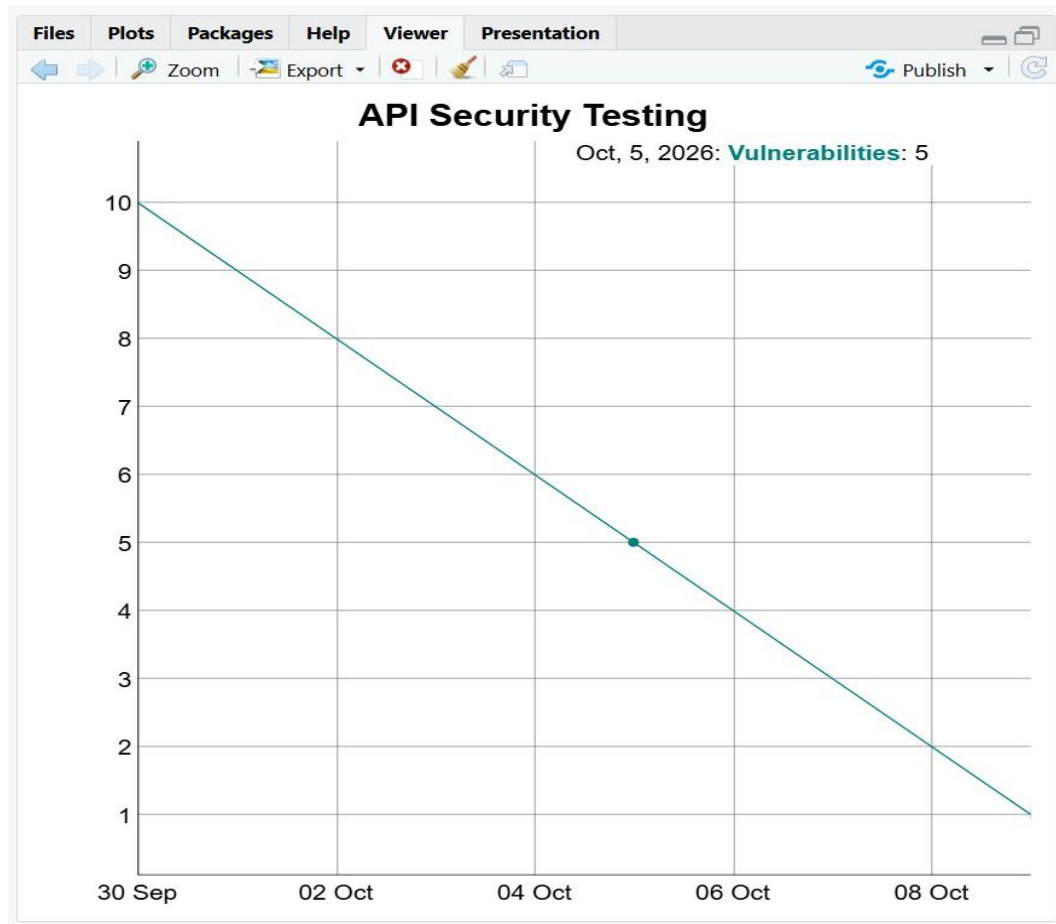
```
  Vulnerabilities = c(10,9,8,7,6,5,4,3,2,1) ),
```

```
order.by=as.Date("2026-09-30")+0:9)
```

```
dygraph(x, main="API Security Testing")
```

Output:

EXPERIMENT 23



Web Application Security Vulnerability Simulation

Type: Dynamic -> dygraph

----- x <-

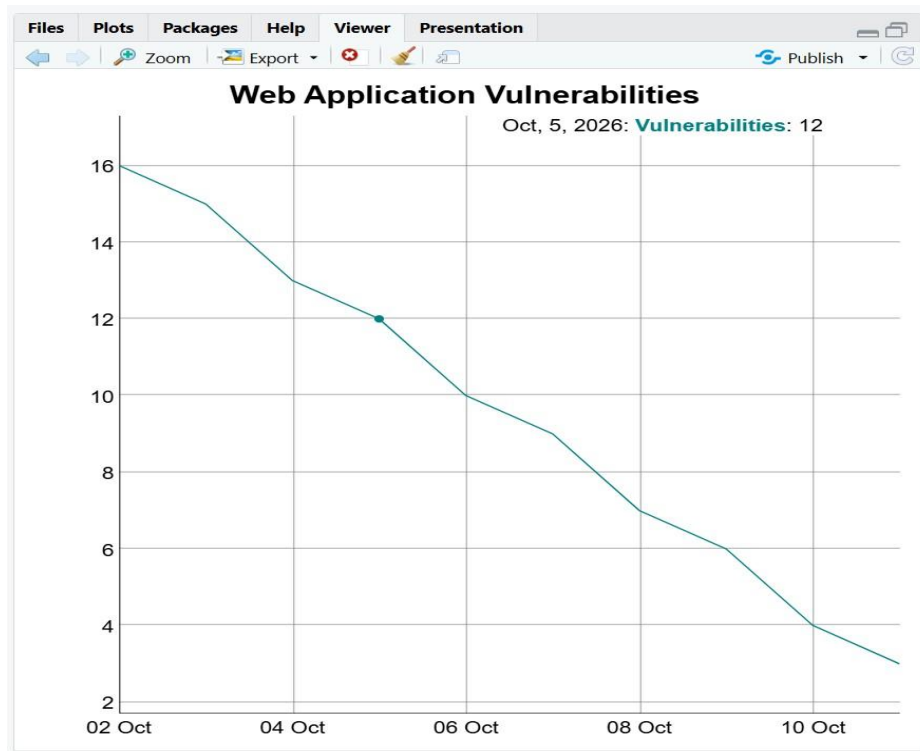
xts(cbind(

Vulnerabilities = c(16,15,13,12,10,9,7,6,4,3)),

order.by=as.Date("2026-10-02")+0:9) dygraph(x,

main="Web Application Vulnerabilities") Output:

EXPERIMENT 24



Security Policy Enforcement using Rule-Based Simulation

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

Allowed = c(8,9,10,11,12,13,14,15,16,17),

Blocked = c(7,6,6,5,4,4,3,2,2,1)),

order.by=as.Date("2026-10-05")+0:9) dygraph(x,

main="Rule-Based Security Policy") Output:

EXPERIMENT 25



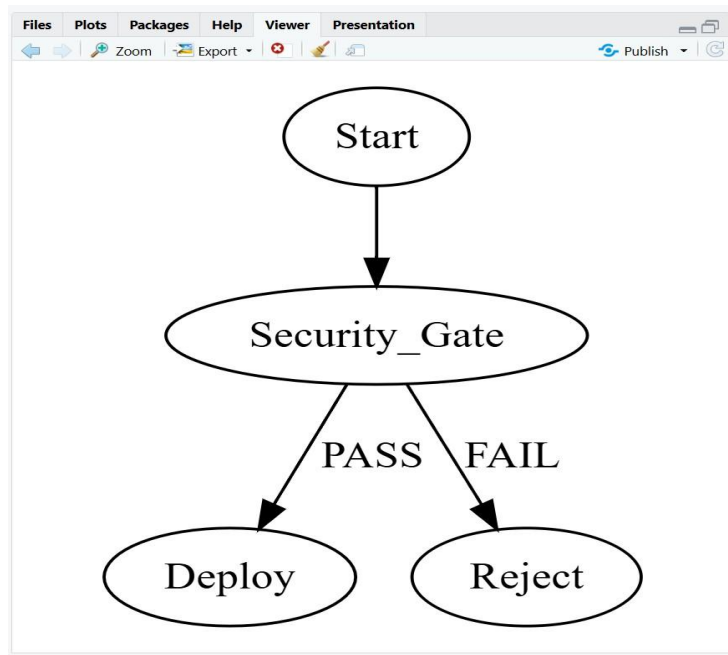
DevSecOps Security Gate Modelling and Decision Simulation

Type: Static -> DiagrammerR

```
grViz(" digraph G {  
  Start -> Security_Gate  
  Security_Gate -> Deploy [label='PASS']  
  Security_Gate -> Reject [label='FAIL']  
}  
")
```

Output:

EXPERIMENT 26



Security Metrics Generation and KPI Analysis

Type: Dynamic -> dygraph

----- x <-

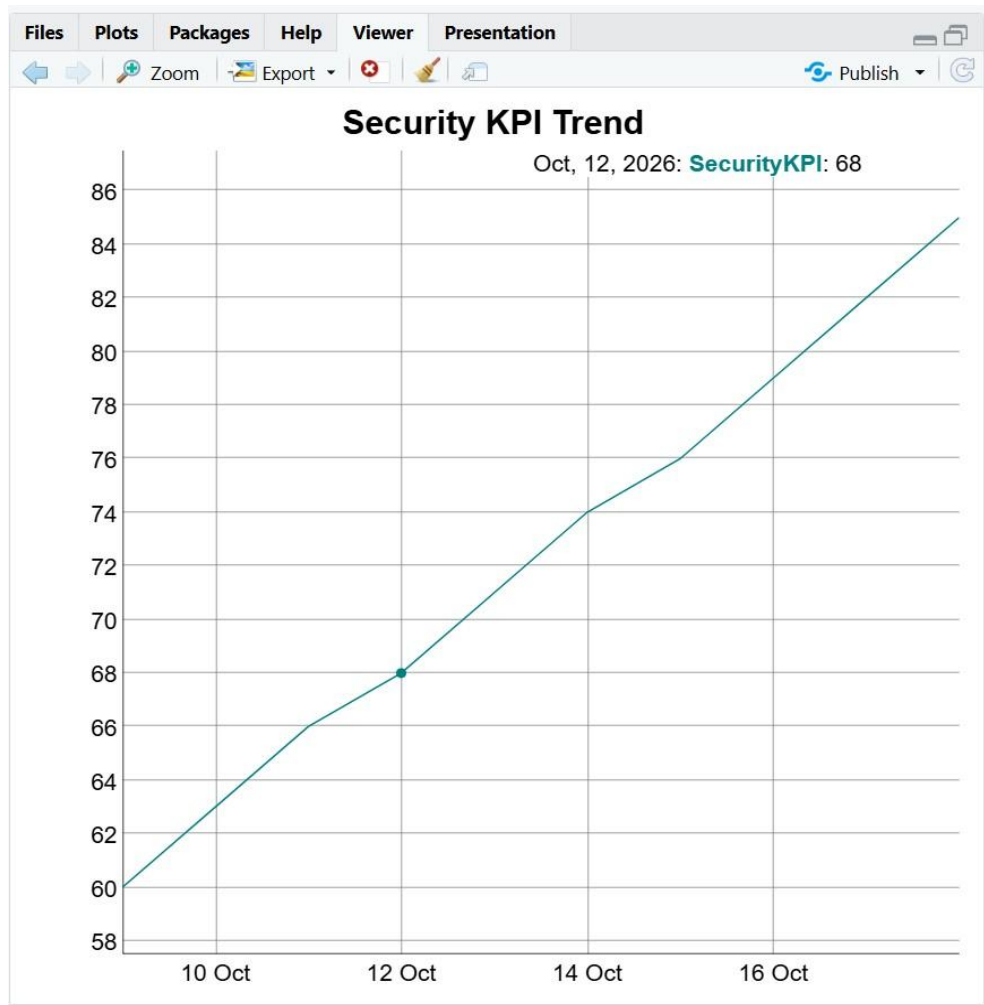
xts(cbind(

SecurityKPI = c(60,63,66,68,71,74,76,79,82,85)),

order.by=as.Date("2026-10-09")+0:9) dygraph(x,

main="Security KPI Trend") Output:

EXPERIMENT 27



Continuous Security Monitoring and Vulnerability Detection Simulation

Type: Dynamic -> dygraph

----- x <-

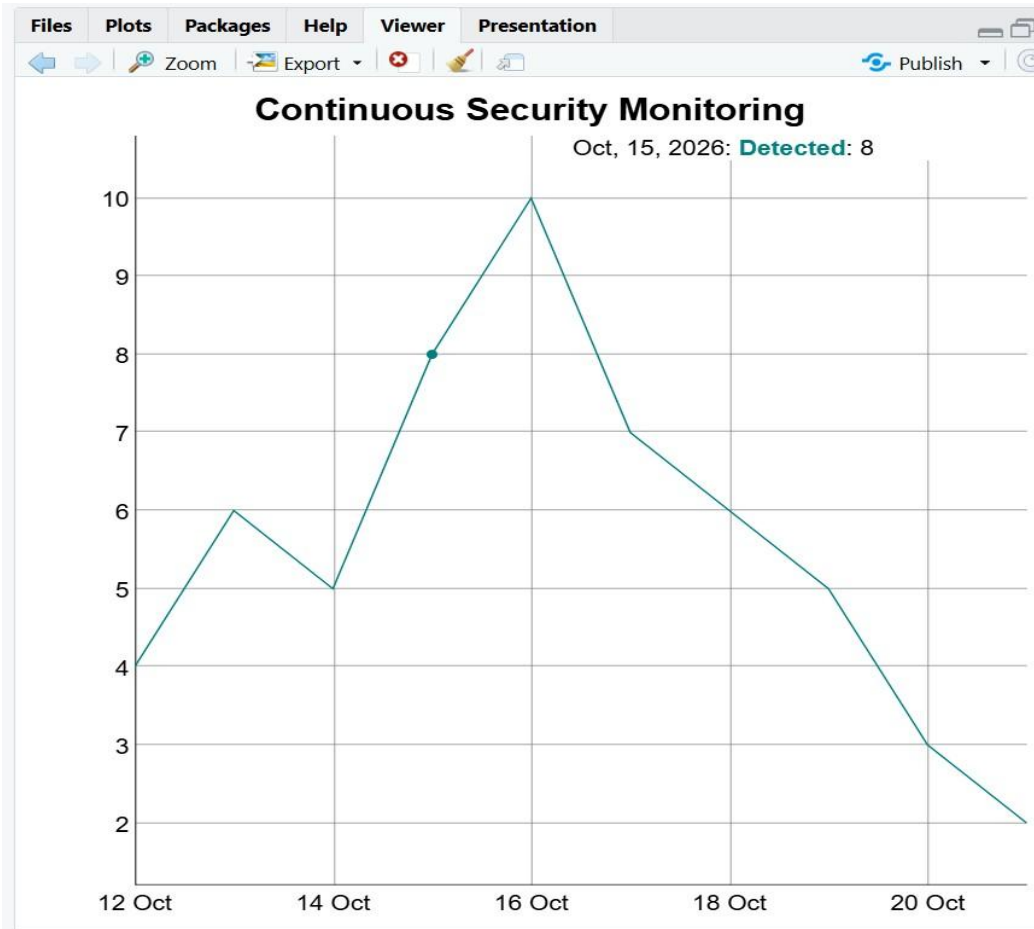
xts(cbind(

Detected = c(4,6,5,8,10,7,6,5,3,2)),

order.by=as.Date("2026-10-12")+0:9) dygraph(x,

main="Continuous Security Monitoring") Output:

EXPERIMENT 28



Security Event Correlation and Attack Pattern Analysis

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

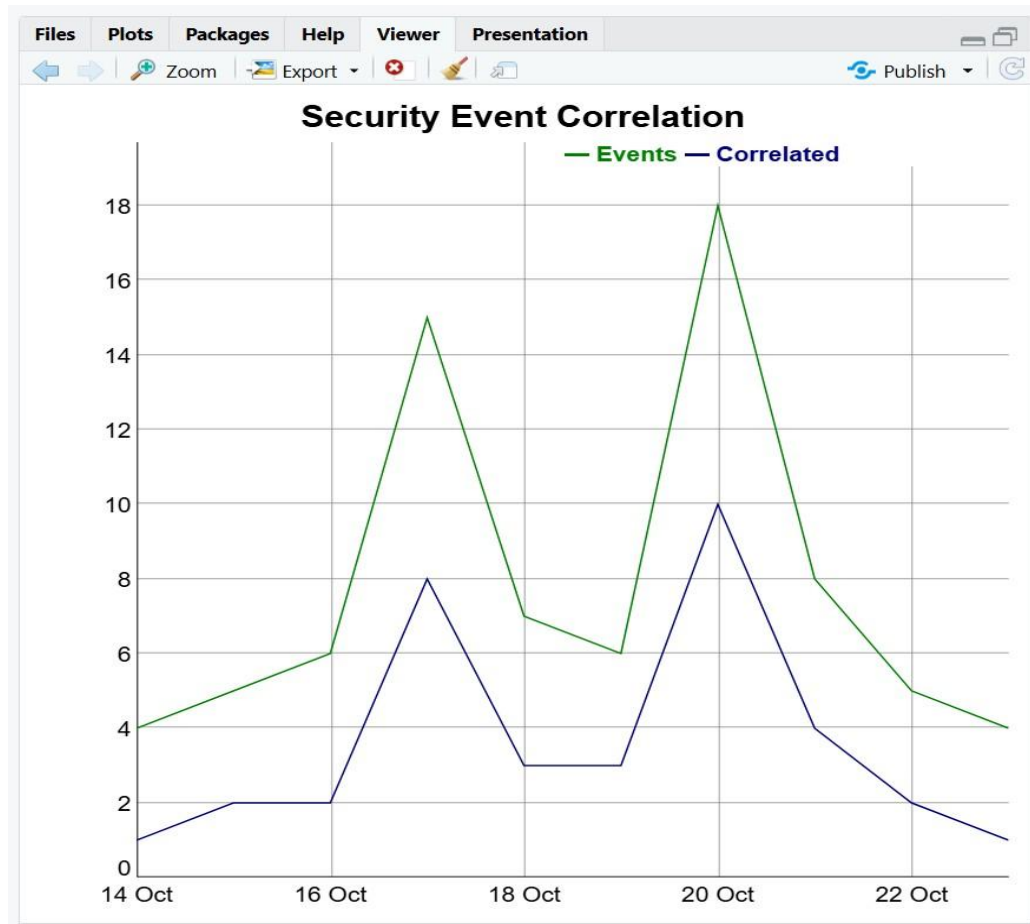
Events = c(4,5,6,15,7,6,18,8,5,4),

Correlated = c(1,2,2,8,3,3,10,4,2,1)),

order.by=as.Date("2026-10-14")+0:9) dygraph(x,

main="Security Event Correlation") Output:

EXPERIMENT 29



Intrusion Detection and Anomaly Detection Simulation

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

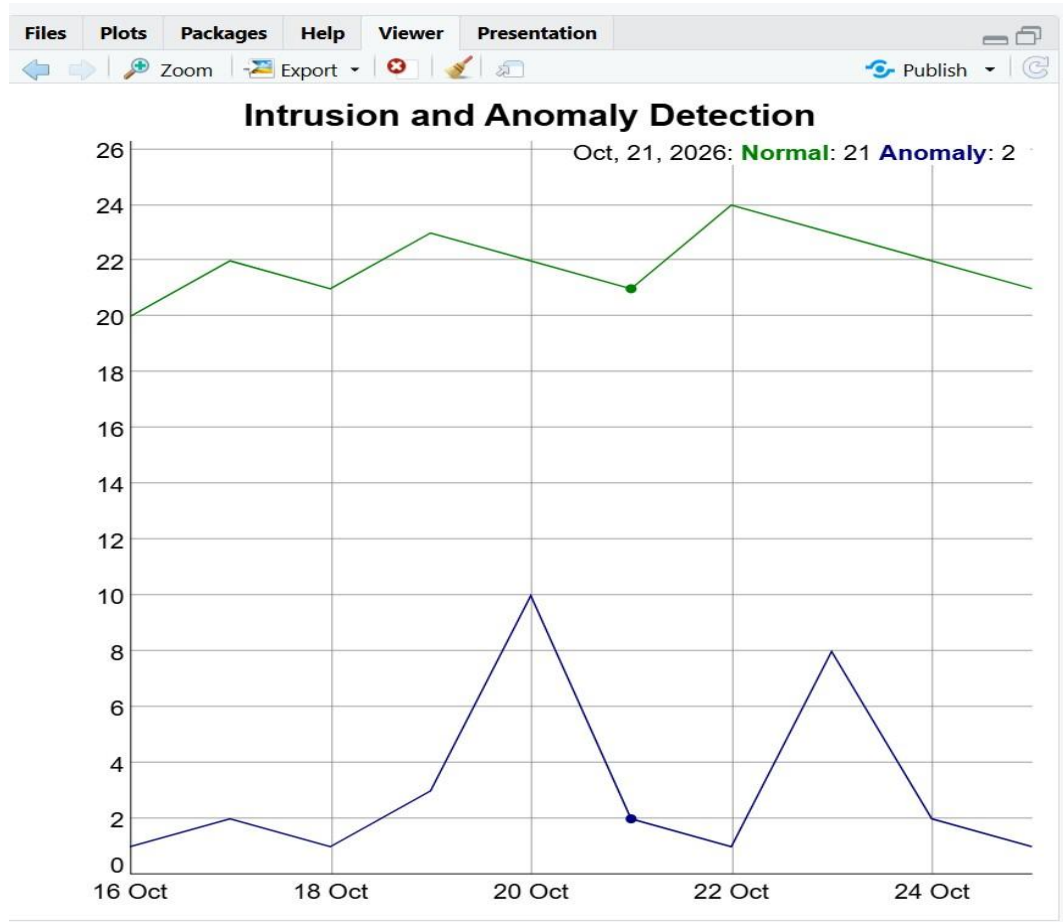
Normal = c(20,22,21,23,22,21,24,23,22,21),

Anomaly = c(1,2,1,3,10,2,1,8,2,1)),

order.by=as.Date("2026-10-16")+0:9) dygraph(x,

main="Intrusion and Anomaly Detection") Output:

EXPERIMENT 30



Security Incident Classification and Severity Analysis

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

Low = c(8,9,10,11,12,13,14,15,16,17),

High = c(6,6,5,5,4,4,3,3,2,2),

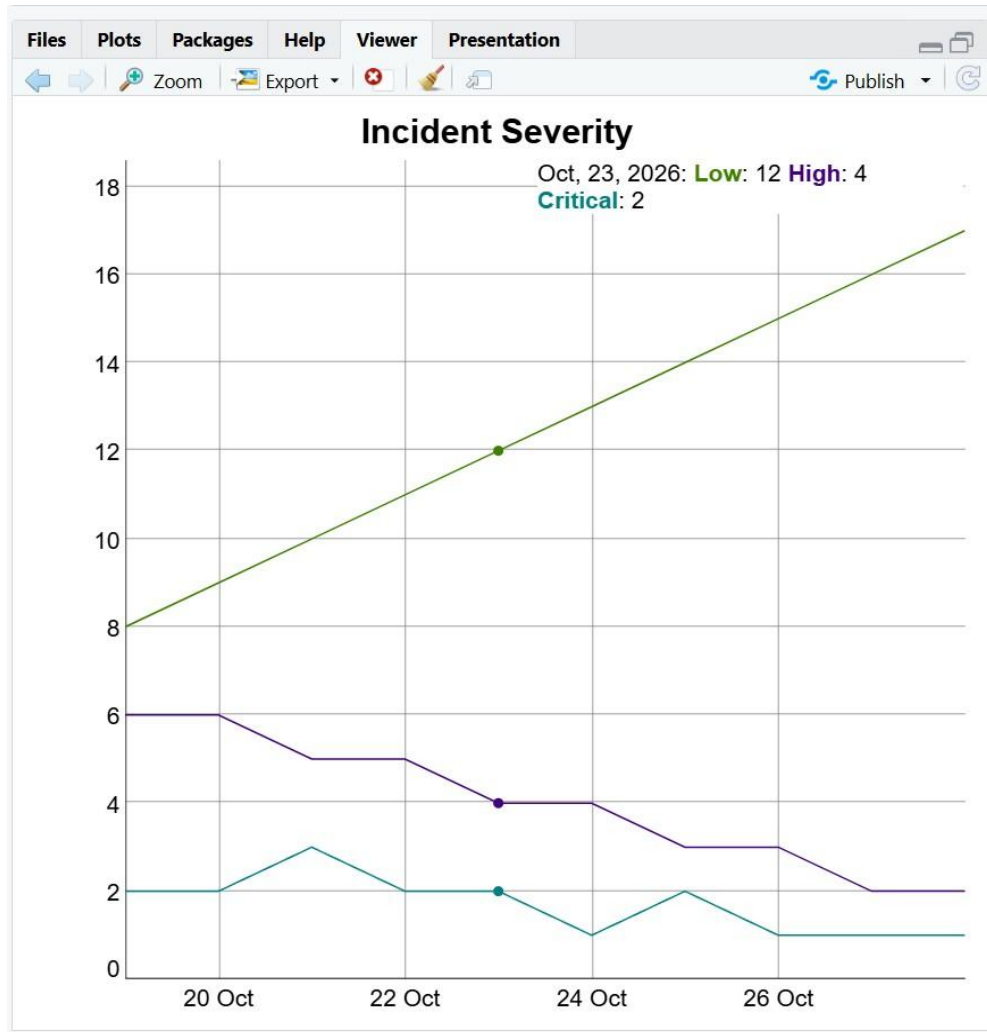
Critical = c(2,2,3,2,2,1,2,1,1,1)),

order.by=as.Date("2026-10-19")+0:9)

dygraph(x, main="Incident Severity")

Output:

EXPERIMENT 31



Incident Response Workflow Modelling and Simulation

Type: Static -> DiagrammerR

```
grViz(" digraph G { rankdir=LR
```

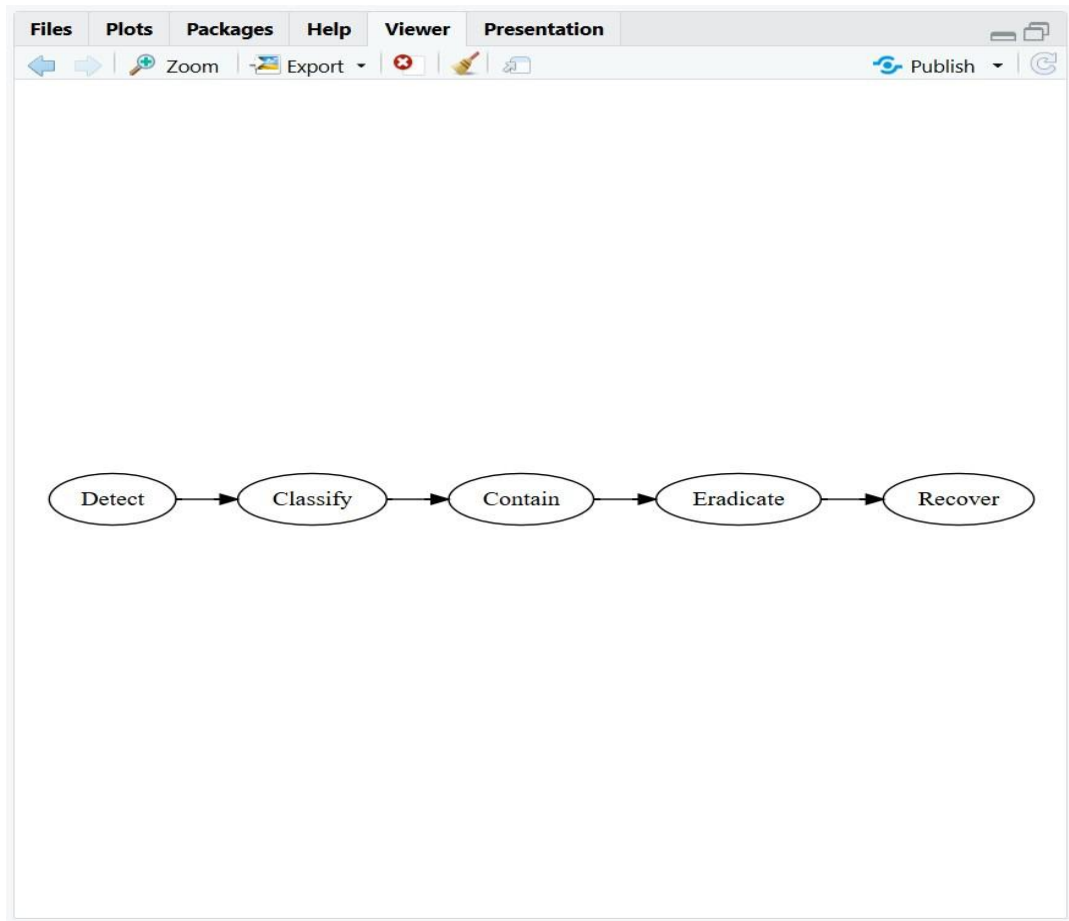
```
    Detect -> Classify -> Contain -> Eradicate -> Recover
```

```
}
```

```
")
```

```
Output;
```

EXPERIMENT 32



Security Incident Timeline Reconstruction and Root Cause Analysis

Type: Dynamic -> dygraph

----- x <-

```
xts(cbind(
```

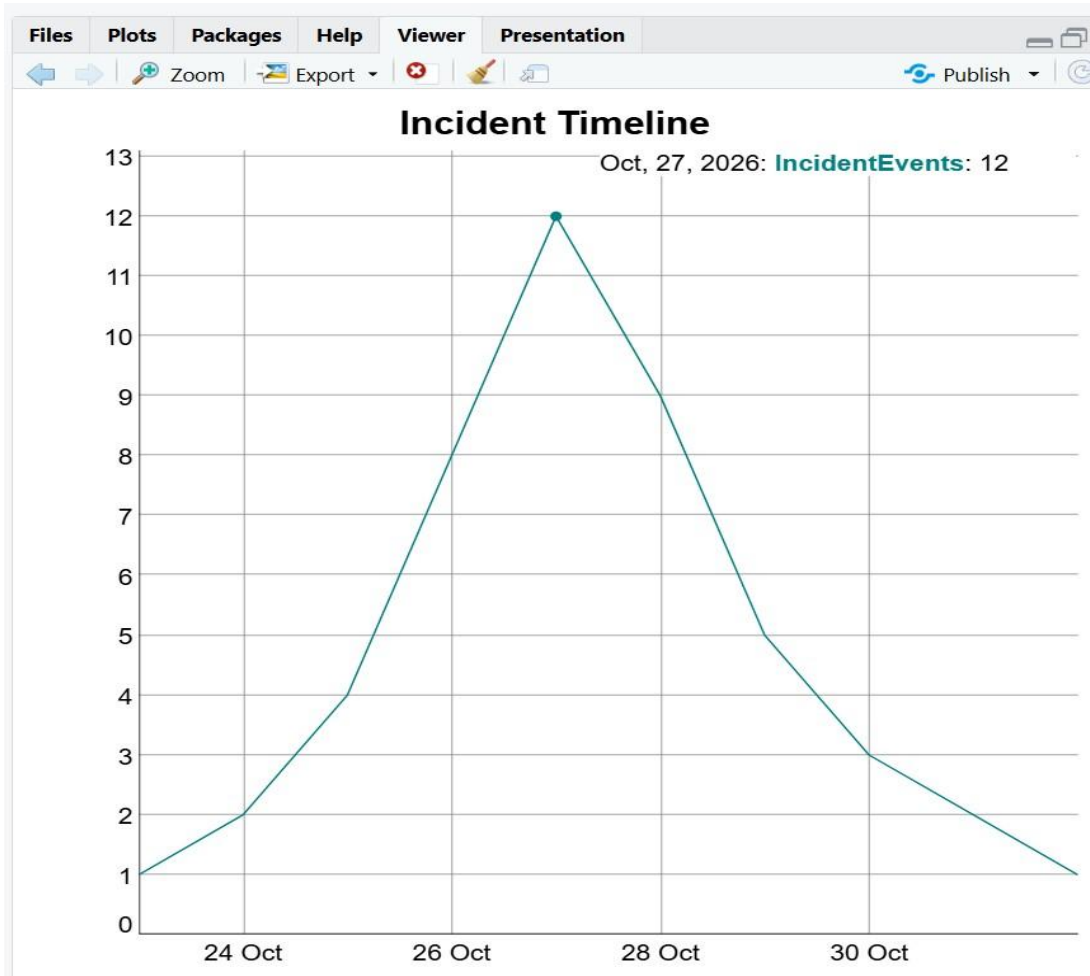
```
  IncidentEvents = c(1,2,4,8,12,9,5,3,2,1) ),
```

```
  order.by=as.Date("2026-10-23")+0:9)
```

```
  dygraph(x, main="Incident Timeline")
```

Output:

EXPERIMENT 33



MTTD and MTTR Analysis

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

MTTD = c(20,18,17,15,14,12,11,10,9,8),

MTTR = c(40,38,35,33,30,28,25,22,20,18)),

order.by=as.Date("2026-10-26")+0:9)

dygraph(x, main="MTTD and MTTR") Output:

EXPERIMENT 34



Security Compliance Monitoring and Governance Simulation

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

Compliance = c(70,72,75,77,80,82,84,87,90,92)),

order.by=as.Date("2026-10-28")+0:9) dygraph(x,

main="Security Compliance") Output:

EXPERIMENT 35



DevSecOps Risk-Based Decision Making using Synthetic Data

Type: Dynamic -> dygraph

----- x <-

xts(cbind(

Risk = c(80,75,70,65,60,55,50,45,40,35)),

order.by=as.Date("2026-10-30")+0:9) dygraph(x,

main="Risk-Based Decision Making") Output:

EXPERIMENT 36



Monte Carlo Simulation for DevSecOps Security Risk Analysis

Type: Dynamic -> dygraph

```
set.seed(1) risk <- rnorm(100, mean=50, sd=10)
```

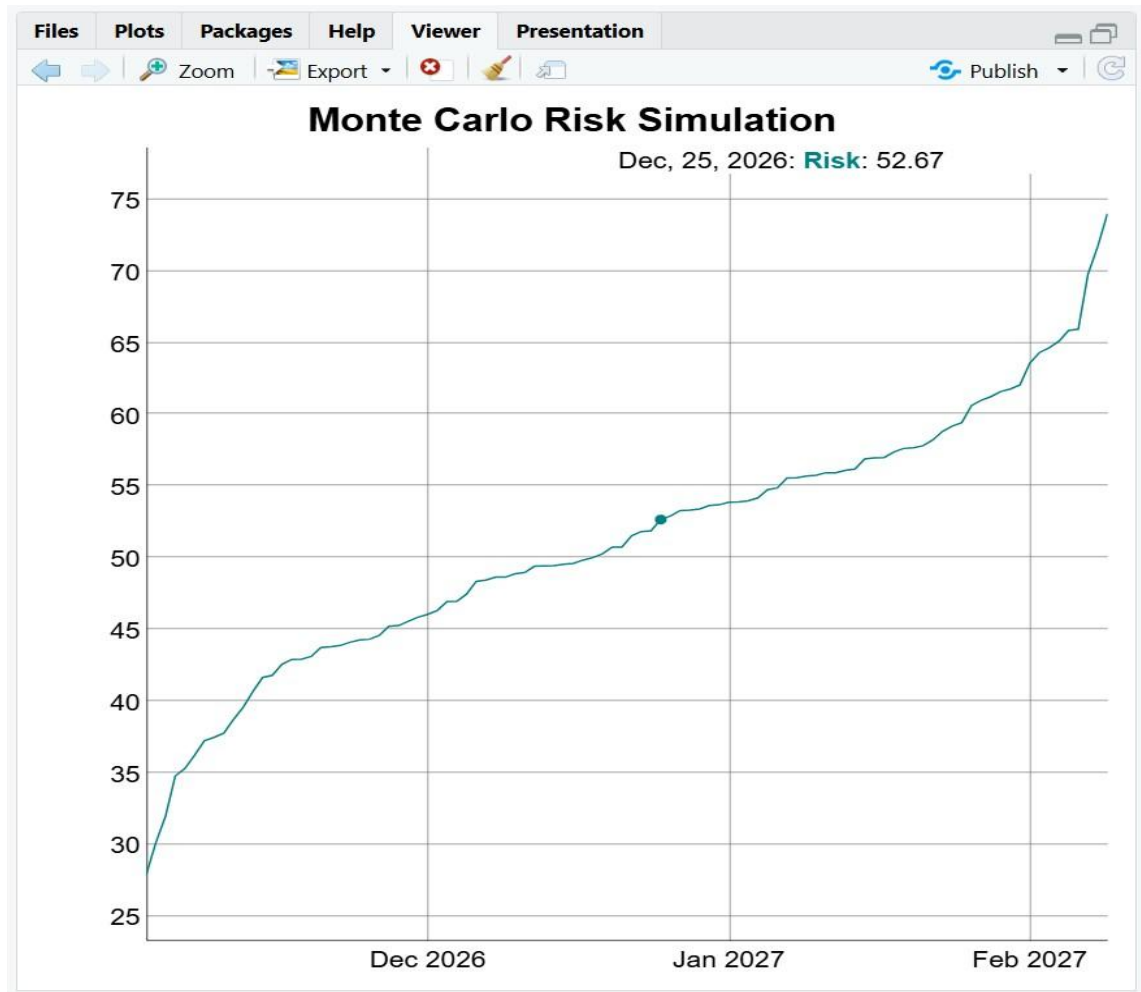
```
x <- xts( cbind(Risk=sort(risk)),
```

```
order.by=as.Date("2026-11-02")+0:99
```

```
)
```

dygraph(x, main="Monte Carlo Risk Simulation") Output:

EXPERIMENT 37



```
# -----
```

```
# EXPERIMENT 38
```

```
# Comparative Evaluation of Secure and Insecure CI/CD Pipelines
```

```
# Type: Dynamic -> dygraph
```

```
# ----- x <-
```

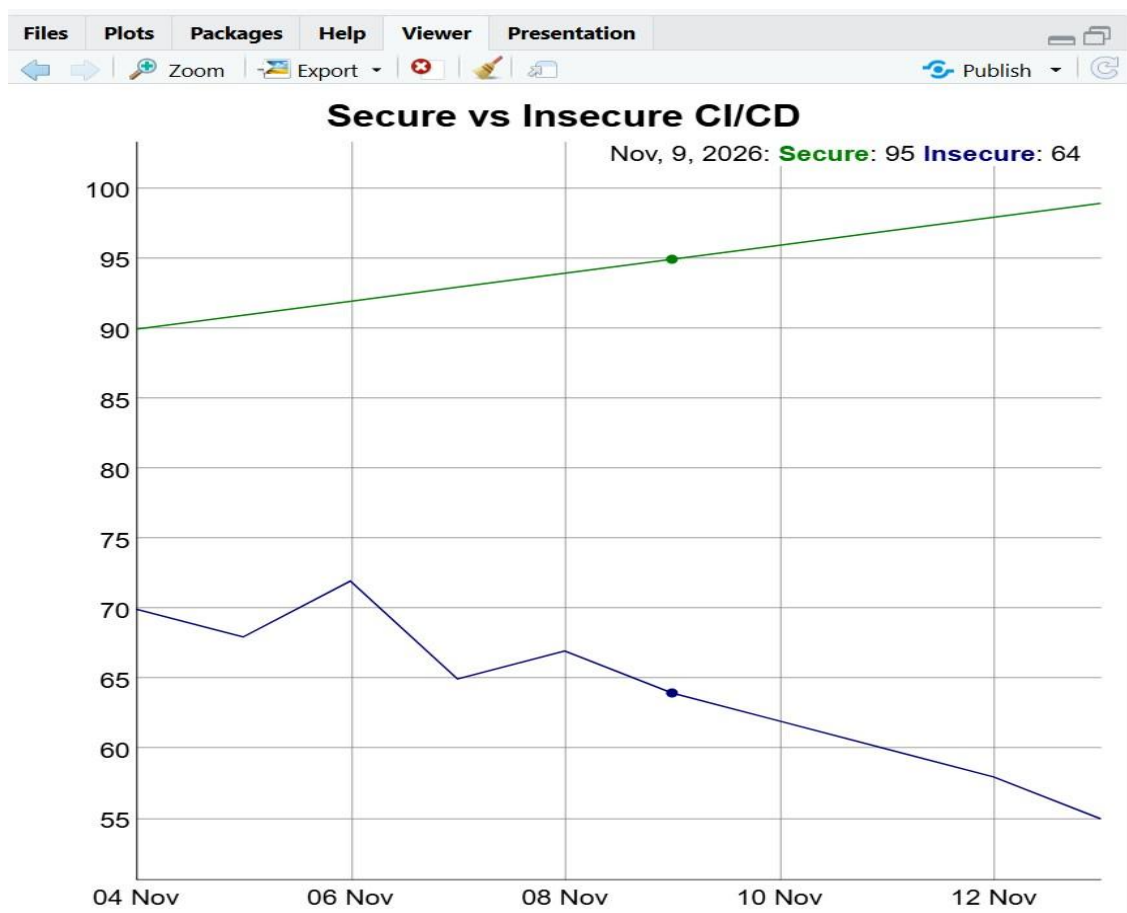
```
xts(cbind(
```

```
  Secure = c(90,91,92,93,94,95,96,97,98,99),
```

```
  Insecure = c(70,68,72,65,67,64,62,60,58,55) ),
```

```
order.by=as.Date("2026-11-04")+0:9) dygraph(x,
```

```
main="Secure vs Insecure CI/CD") Output:
```



```
# -----
```

```
# EXPERIMENT 39
```

```
# Integrated DevSecOps Security Analytics and Reporting
```

```
# Type: Dynamic -> dygraph
```

```
# ----- x <-
```

```
xts(cbind(
```

```
  SecurityScore = c(60,64,68,70,73,76,79,82,85,88),
```

```
  Vulnerabilities = c(20,18,17,15,13,11,9,7,5,4) ),
```

```
order.by=as.Date("2026-11-06")+0:9) dygraph(x,
```

```
main="Integrated Security Analytics")
```



```
# -----
```



```
# EXPERIMENT 40
```

```
# End-to-End DevSecOps Pipeline Security Simulation and Evaluation
```

```
# Type: Dynamic -> dygraph
```

```
# ----- x <-
```

```
xts(cbind(
```

```
  Build = c(70,72,75,78,80,82,84,86,88,90),
```

```
  Security = c(60,64,68,72,75,78,81,84,87,90),
```

```
  Deployment = c(55,60,65,68,72,76,80,84,87,91) ),
```

```
order.by=as.Date("2026-11-09")+0:9) dygraph(x,
```

```
main="End-to-End DevSecOps Pipeline") Output:
```

